

Objective

The objective of Day 10 is to develop a practical **Contact Book application using Python dictionaries**. The application demonstrates how key-value data can be stored and managed efficiently. Users can add new contacts, search for existing contacts, update contact information, and delete contacts. The project provides practical experience with dictionaries, functions, conditional statements, validation, and CRUD operations.

Key Features

- Add new contacts.
- View all saved contacts.
- Search contacts using a phone number.
- Update existing contact information.
- Delete contacts.
- Use the phone number as the dictionary key.
- Store contact name and email as values.
- Prevent duplicate phone numbers.
- Validate phone-number input.
- Interactive Streamlit interface.
- Maintain contacts using Streamlit session state.

Technologies Used

The project uses **Python and Streamlit**. A Python dictionary is used as the main data structure, where the phone number acts as the key and the contact details such as name and email are stored as values. Python functions are used to perform the different contact-management operations, while conditional statements provide validation and error handling. Streamlit is used to create the interactive localhost interface **Working Principle**

The application stores contact information in a Python dictionary. Each phone number is used as a unique key, while the corresponding contact name and email are stored as values. When a user adds a contact, the program first checks whether the phone number already exists. During a search, the phone number is used to retrieve the corresponding dictionary value. Existing contact information can be updated, while unwanted contacts can be removed using the delete operation.

Learning Outcomes

This project helps develop a practical understanding of **Python dictionaries, key-value pairs, dictionary methods, functions, CRUD operations, input validation, and Streamlit application development**. It also demonstrates why dictionaries are useful when information needs to be retrieved using a unique key.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY10>

```
app.py X
app.py > ...
1  import streamlit as st
2
3  st.set_page_config(
4      page_title="Contact Book",
5      page_icon="📞",
6      layout="centered"
7  )
8  if "contacts" not in st.session_state:
9      st.session_state.contacts = {
10         "9876543210": {
11             "name": "Rahul",
12             "email": "rahul@example.com"
13         },
14         "9123456780": {
15             "name": "Priya",
16             "email": "priya@example.com"
17         }
18     }
19
20 def add_contact(phone, name, email):
21     """Add a new contact."""
22
23     if phone in st.session_state.contacts:
24         return False, "A contact with this phone number already exists."
25
26     st.session_state.contacts[phone] = {
27         "name": name,
28         "email": email
29     }
30
31     return True, "Contact added successfully."
32
33
34 def search_contact(phone):
35     """Search for a contact using phone number."""
36
37     return st.session_state.contacts.get(phone)
38
39
40 def update_contact(phone, name, email):
41     """Update an existing contact."""
42
43     if phone not in st.session_state.contacts:
44         return False
45
46     st.session_state.contacts[phone] = {
47         "name": name,
```

```
def update_contact(phone, name, email):
    "email": email
    }

    return True

def delete_contact(phone):
    """Delete an existing contact."""

    if phone not in st.session_state.contacts:
        return False

    del st.session_state.contacts[phone]
    return True
st.markdown("""
<style>
    .title {
        text-align: center;
        font-size: 42px;
        font-weight: bold;
        margin-bottom: 5px;
    }

    .subtitle {
        text-align: center;
        font-size: 18px;
        margin-bottom: 25px;
    }

    .contact-card {
        padding: 15px;
        margin: 10px 0;
        border-radius: 10px;
        border: 1px solid #dddddd;
        font-size: 16px;
    }
</style>
""", unsafe_allow_html=True)

st.markdown(
    '<div class="title"> 📞 Contact Book</div>',
    unsafe_allow_html=True
)

st.markdown(
    '<div class="subtitle">'
```

OUTPUT: http://localhost:8501

 **Contact Menu**

Choose an operation:

☐  View Contacts

☐  Add Contact

☐  Search Contact

☒  Update Contact

☐  Delete Contact

 **Contact Menu**

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Contact Book

Manage your contacts using Python dictionaries



Update Contact

Enter phone number of contact:

7843092176

Contact found.

New Name

sharvari

New Email

tfeugjhd@gmail.com

Update Contact



Contact Book

Manage your contacts using Python dictionaries



Search Contact

Enter phone number:

7843092176

Search

✔ Contact found!

Name: sharvari

Phone: 7843092176

Email: tfeugjhd@gmail.com

Contact Menu

Choose an operation:

- ☐  View Contacts
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- ☐  Search Contact
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- ☒  Delete Contact



Contact Book

Manage your contacts using Python dictionaries

+ Add New Contact

Contact Name

sharvari

Phone Number

7843092176

Email Address

tfeugjhd@gmail.com

Add Contact

✓ Contact added successfully.



Contact Book

Manage your contacts using Python dictionaries



All Contacts

Total Contacts: 2

 Rahul

 9876543210

 rahul@example.com

 Priya

 9123456780

 priya@example.com



Contact Book

Manage your contacts using Python dictionaries



Delete Contact

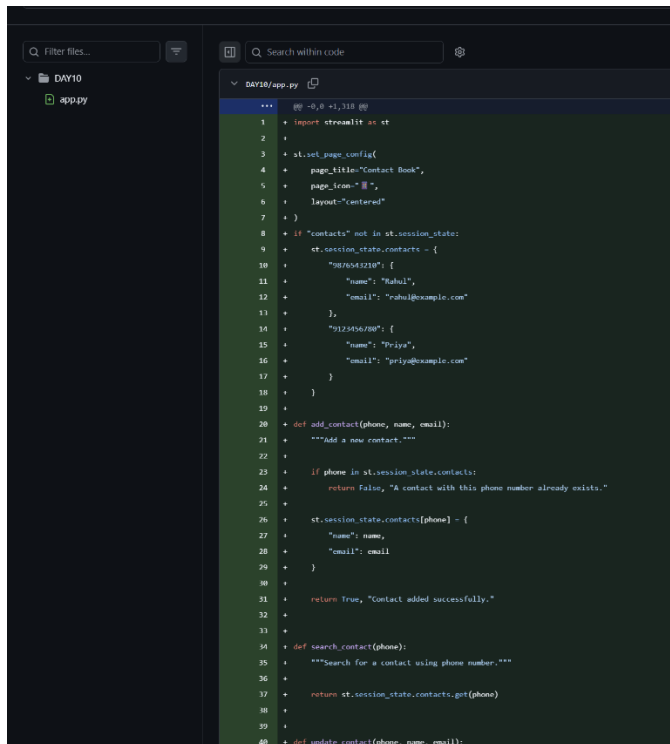
Enter phone number:

9123456780

Delete Contact

✗ Contact not found.

Github:<https://github.com/shambhavi0804/VEDA-internship/commit/5da89da6b468b40d94a666ad5edfdd5410c60bbf>



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Linkedin:<https://lnkd.in/p/d7J3cYim>

